

สิ่งที่ไม่ใช่ฟังก์ชัน

Discontinuity $\rightarrow$ Removable: $\lim_{x \rightarrow a} f(x)$ exists but not equal to $f(a)$ (or $f(a)$ DNE)

Jump: $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$

Infinite: $\lim_{x \rightarrow a} f(x) / \lim_{x \rightarrow a} f(x) = \infty / -\infty$

Oscillates: oscillate infinitely with amplitude $\geq$ fixed positive number

Asymptotes $\rightarrow \lim_{x \rightarrow \infty / -\infty} f(x) = L$ (Horizontal)

$\rightarrow$ Infinite discontinuity (Vertical)

$\rightarrow \lim_{x \rightarrow \infty / -\infty} f(x) = Ax + B$; A, B const

Marginal cost = $c'(x) = \frac{dc}{dx}$

Cost $\uparrow$ products $\uparrow$

Linearization: $f(x) \approx f(a) + (x-a) \cdot f'(a)$

o/o : $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = 0 \rightarrow f(x) = o(g(x))$ as $x \rightarrow c$

$\exists c > 0$ s.t. $|f(x)| \leq M(g(x))$; $x \approx c$, M const $\rightarrow f(x) \sim O(g(x))$

$\Delta f \rightarrow$ finite $df \rightarrow$ infinitesimal tangent

Critical pts: $f'(x) = 0$ or DNE / Boundary (Closed interval) CP must be interior point of domain (Local max/min) $\rightarrow$ ระวังที่ขอบด้วย

Inflection pts: concavity changes (from $+$ to $-$ / $-$ to $+$) ($f''(x)$ doesn't need to be 0)

If $f(x)$ continuous on $[a, b] \rightarrow$ exist local max/min (ระวัง closed interval)

Rolle's: $f(a) = f(b) \rightarrow \exists c \in [a, b]$ s.t. $f'(c) = 0$

MVT: $\exists c \in [a, b]$ s.t. $\frac{f(b) - f(a)}{b - a} = f'(c)$

Newton's method: $X_{n+1} = X_n - \frac{f(X_n)}{f'(X_n)}$

Integrable: $\rightarrow$ Riemann sum $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x_k$ exists / independent of choice c_i

$\downarrow$ At most finitely jump/removable discontinuity

$$\sqrt{x^2} = -x \text{ if } x < 0$$

ระวังที่ f, g, h fucked up ที่ขอบที่ปิดด้วย

ถ้า $f(x)$ ที่ boundary ด้วย แล้วเป็น local max/min หรือถ้าไม่ระวัง CP ที่คำนวณ (4 optimize)

(sin/cos)

$$0 \times \text{Bounded} = 0$$

ระวังที่ $f'(x), f''(x) > < = 0$ แล้วที่ขอบด้วย

5. (Midterm)

$$m(x) = c'(x) = 150 + x \quad \left. \begin{array}{l} \alpha(x) = \frac{2000}{x} + 150 + 0.5x \\ \alpha(x) = c(x)/x \end{array} \right\} \frac{2000}{x} + 0.5x = x \rightarrow x^2 = 40000 \\ x = 200$$

$$\rightarrow \alpha'(x) = \frac{xc'(x) - c(x)}{x^2} = c'(x) - \alpha(x)$$

$$\text{At } x_0, \alpha'(x) = 0 \rightarrow \alpha(x) = c'(x) = m(x)$$

Mean Value Theorem for Definite Integrals

→ Suppose $f(x)$ is continuous on $[a, b]$, then $\frac{1}{b-a} \int_a^b f(x) dx = f(c)$ for some $c \in [a, b]$

Fundamental Theorem Part 1

Assume $F(x)$ contin. in $[a, b]$, define $F(x) = \int_a^x f(t) dt$; $x \in [a, b]$

$$\rightarrow F(a) = 0, F(b) = \int_a^b f(t) dt$$

Then $F(x)$ is differentiable on $[a, b]$ and $F'(x) = f(x) \forall x \in [a, b]$ (F is antiderivative of f)

Suppose f is contin on $[a, b]$ and $F(x)$ is an antiderivative of $f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a) = F(x) \Big|_a^b$$

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(t) dt$$

$$\text{Area under curves} = \int_a^b (f(x) - g(x)) dx$$

$$g(x) \leq f(x) \text{ on } [a, b]$$

$$V = \int_a^b A(x) dx$$

rotate about X -axis

$$V = \int_a^b A(x) dx = \int_a^b \pi (f(x))^2 dx$$

Hollow cylinder

$$V = \int_a^b \pi (f^2(x) - g^2(x)) dx$$

$$\text{Arc length: } L = \int_a^b \sqrt{1 + (f'(x))^2} dx = \int_a^b \sqrt{(dx)^2 + (dy)^2} = \int ds$$

$$\text{Surface Area } A = \int_a^b 2\pi f(x) \sqrt{(dx)^2 + (dy)^2}$$

$$W = \int_a^b F(x) dx \quad (\text{Work})$$

$$\text{Fluid force} = \int_a^b w \cdot (d-y) \cdot \underbrace{\ell(y) dy}_{\substack{\text{A (strip area at depth } y) \\ \text{depth from surface}}}$$

$$\frac{dx}{dy} \Big|_{y=b} = \frac{1}{\frac{dy}{dx} \Big|_{x=a}}$$

$$(f^{-1}(b))' = \frac{1}{f'(a)} \quad \text{when } f(a) = b$$

$$(\log_a x)' = \left(\frac{\ln x}{\ln a} \right)' = \frac{1}{x \ln a} \quad \frac{d}{dx} (\log_a u) = \frac{1}{\ln a} \cdot \frac{1}{u} \frac{du}{dx}$$

DO NOT burn your midnight oil

After all, tomorrow is another day

... a well to increase
 ... have determined that
 the pressure needed for
 ... shown here. The
 ... a vertical 10-in
 ... diameter and 7.5 in
 ... ground. The pump
 ... the current hour, how
 ... (Include the time it
 ... the 9000 ft³/m³)



Problem

1. (Xiv)

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \right)$$

$$\lim_{n \rightarrow \infty} \left[\frac{1}{n} \left(\frac{1}{1+\frac{1}{n}} + \frac{1}{1+\frac{2}{n}} + \dots + \frac{1}{1+\frac{n}{n}} \right) \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{1+\frac{k}{n}}$$

$$= \int_a^b f(x) dx$$

$x = \frac{1}{n} \times k$
 $x_k = \frac{k}{n}$
 $\int_1^2 \frac{1}{1+x} dx$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \cdot \frac{1}{1+\frac{k}{n}} \Delta x$$

$$\int_1^2 \frac{1}{1+x} dx$$



$$\int_{423}^{2025} u \, du$$

$$\frac{y}{\theta}$$

Sweep the floor

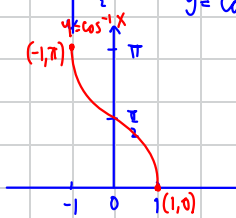
$y = \sin^{-1}(x)$, $x \in [-1, 1]$ if and only if $\sin y = x$ and $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$
 $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}(x) + C$

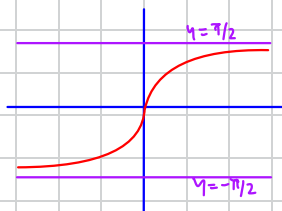
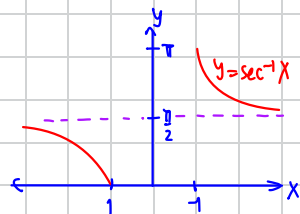
$y = \cos^{-1}(x)$, $x \in [-1, 1]$ if and only if $y \in [0, \pi]$, $\cos y = x$

$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
 $\frac{d}{dx}(\cos^{-1} x) = \frac{d}{dx}(\frac{\pi}{2} - \arcsin x) = -\frac{1}{\sqrt{1-x^2}}$

$x \in (0, \frac{\pi}{2})$ $\cos(0, \frac{\pi}{2})$
 $\frac{\sin x - \sin 0}{x - 0} = \cos c < 1$
 $\therefore \sin x < x$
 $\frac{\tan x - \tan 0}{x - 0} = \sec^2 c > 1$
 $\therefore \tan x > x$



$(-\infty, \infty) = \text{Dom}(\tan^{-1})$
 $(-\frac{\pi}{2}, \frac{\pi}{2}) = \text{Range}(\tan^{-1})$



$(\tan^{-1} x)' = \frac{1}{1+x^2}$
 $\int \frac{1}{1+x^2} dx = \tan^{-1}(x) + C$

$y = \sec^{-1} x, x \in (-\infty, -1] \cup [1, \infty) \iff y \in [0, \pi] \setminus \{\frac{\pi}{2}\}$

$\int \frac{dx}{|x|\sqrt{x^2-1}} = \sec^{-1}(x) + C$
 $\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1}|x| + C$

$u = f(x), v = g(x)$
 $du = f'(x)dx, dv = g'(x)dx$

$\int u dv = uv - \int v du$

$\int (\sin x)^m (\cos x)^n dx$

m is odd ($m=2k+1$) $\rightarrow \int \sin^m x \cdot \cos^n x dx = \int (1-\cos^2 x)^k \cdot \cos^n x \cdot d(\cos x)$

n is odd ($n=2k+1$) $\rightarrow \int \sin^m x \cdot \cos^n x dx = \int \sin^m x \cdot (1-\sin^2 x)^k \cdot d(\sin x)$

m, n even ($m=2k, n=2l$) $\rightarrow \int \sin^m x \cdot \cos^n x dx = \int (\sin^2 x)^k \cdot (\cos^2 x)^l dx = \int (\frac{1-\cos 2x}{2})^k (\frac{1+\cos 2x}{2})^l dx$

$\int \sin mx \cos nx dx = \frac{1}{2} \int [\sin(mx+nx) + \sin(mx-nx)] dx$
 $= \frac{1}{2} \left[\frac{-\cos(mx+nx)}{m+n} - \frac{-\cos(mx-nx)}{m-n} \right] + C$

$\int \sin mx \sin nx dx = \frac{1}{2} \int [\cos(mx-nx) - \cos(mx+nx)] dx$
 $= \frac{1}{2} \left[\frac{\sin(mx-nx)}{m-n} - \frac{\sin(mx+nx)}{m+n} \right] + C$

$\int \cos mx \cos nx dx = \frac{1}{2} \int [\cos(mx-nx) + \cos(mx+nx)] dx$
 $= \frac{1}{2} \left[\frac{\sin(mx-nx)}{m-n} + \frac{\sin(mx+nx)}{m+n} \right] + C$

$\int \tan^m x \sec^n x dx$

$\uparrow n$ even $n=2k \rightarrow \int \tan^m x (\sec^{2k-2} x) (\sec^2 x dx)$
 $(\tan x + 1)^{k-1} d(\tan x)$

$2) n=0; m \geq 2 \int \tan^m x dx = \frac{\tan^{m-1} x}{m-1} - \int \tan^{m-2} x dx$

$3) m=0; n \geq 3$
 $\int \sec^n x dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{(n-2)}{n-1} \int \sec^{n-2} x dx$

$4) n=1 \int \tan^{m-1} x \sec x \tan x dx$
 $= \tan^{m-1} x \sec x - (m-1) \int \sec x \cdot \tan^{m-2} x \sec^2 x dx$
 $= \frac{1}{m} (\tan^{m-1} x \sec x - (m-1) \int \sec x \cdot \tan^{m-2} x dx)$

$\tan \rightarrow \ln|\sec|$
 $\sec \rightarrow \ln|\sec + \tan|$
 $\tan^2 \rightarrow \tan - x$
 $\sec^2 \rightarrow \tan$
 $\sec \tan \rightarrow \sec$

Left sum $L_n = \sum_{i=1}^n f(x_{i-1}) \Delta x$
 Right sum $R_n = \sum_{i=1}^n f(x_i) \Delta x$
 Middle sum $M_n = \sum_{i=1}^n f\left(\frac{x_{i-1} + x_i}{2}\right) \Delta x$

Error:
 $E_n^R \leq \frac{(b-a)^2}{2n} \max_{[a,b]} |f'(x)| \rightarrow 0$
 $E_n^L = O\left(\frac{1}{n}\right)$ as $n \rightarrow \infty$

$E_n^M \leq \frac{(b-a)^3}{24n^2} \max_{[a,b]} |f''(x)| = O\left(\frac{1}{n^2}\right)$

$\int_0^{1/2} \frac{\cos t}{t + \sin t} dt$
 $\lim_{t \rightarrow 0^+} \frac{\cos t}{t + \sin t} = \lim_{t \rightarrow 0^+} \frac{t}{t + \sin t} \cdot \lim_{t \rightarrow 0^+} \frac{\cos t}{1} = 1$
 $\therefore \int_0^{1/2} \frac{\cos t}{t + \sin t} dt$ grows same rate w/ $\int_0^{1/2} \frac{1}{t} dt$ which diverges.

Trapezoidal Rule: $T_n = \sum_{i=1}^n \frac{f(x_{i-1}) + f(x_i)}{2} \Delta x = \frac{\Delta x}{2} (y_0 + 2y_1 + 2y_2 + \dots + 2y_{n-1} + y_n)$
 $= \frac{1}{2}(L_n + R_n)$

T_n overestimate if f concave up (M_n underestimate)
 underestimate if f concave down (M_n overestimate)

$|E_n^T| \leq \frac{(b-a)^3}{12n^2} \max_{[a,b]} |f''(x)| = O\left(\frac{1}{n^2}\right)$

Simpson's Rule $S_n = \frac{\Delta x}{3} [y_0 + 4y_1 + 2y_2 + 4y_3 + \dots + 2y_{n-2} + 4y_{n-1} + y_n]$

$|E_n^S| \leq \frac{(b-a)^5}{180n^4} \max_{[a,b]} |f^{(4)}(x)| = O\left(\frac{1}{n^4}\right)$

Type I: $x \rightarrow \infty$ or $-\infty$
 Type II: $y \rightarrow \infty$ or $-\infty$

n large,
 Simpson Best
 Midpoint
 Trapezoid
 Left Right Worst

$\int_a^{\infty} f(x) dx$ as $\lim_{b \rightarrow \infty} \int_a^b f(x) dx$

Suppose $0 \leq f(x) \leq g(x)$, $x \in [a, \infty)$

- If $\int_a^{\infty} g(x) dx$ converges, then $\int_a^{\infty} f(x) dx$ converges
- If $\int_a^{\infty} f(x) dx$ diverges, then $\int_a^{\infty} g(x) dx$ diverges

p -test
 $\int_a^{\infty} \frac{1}{x^p} dx$
 $p > 1$ Diverge
 $p = 1$ Diverge
 $p < 1$ Converge

$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L$; f, g contin on $[a, \infty)$ and > 0

- $\int_a^{\infty} f(x) dx$ and $\int_a^{\infty} g(x) dx$ converge or diverge at the same rate
- If $L = 0$ then if $\int_a^{\infty} g(x) dx$ converges, then $\int_a^{\infty} f(x) dx$ converges

DE: Separable eq. $\frac{dy}{dx} = f(x)g(y)$

$\int \frac{1}{g(y)} dy = \int f(x) dx$

Linear $\frac{dy}{dx} + a(x)y = b(x)$

$v(x) = e^{\int a(x) dx} \rightarrow (v(x) \cdot y)' = b(x)v(x)$
 $\rightarrow y = \frac{1}{v(x)} \int b(x)v(x) dx$

Euler's method: numerical approximation

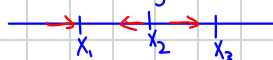
$y = y_0 + f(x, y)(x - x_0)$

$y_1 = y_0 + f(x_0, y_0) \Delta x$

Newton's Cooling Method $\frac{dT}{dt} = -k(H - H_s)$; k const > 0

Solute concentration $\frac{dM}{dt} = \text{rate solute in} - \text{rate solute out}$
 $\text{conc} \times \text{rate liquid in} - \frac{y(t)}{\text{volume}} \times \text{rate liquid out} \rightarrow \text{linear DE}$

Phase line analysis



- Find equilibrium sols.
 - Mark on phase line
 - Draw arrows on each interval
- $\left(\frac{dy}{dt} > 0 \rightarrow \frac{dy}{dt} < 0 \leftarrow\right)$